Artificial Neural Network Architecture for Customer Churn Prediction

Project: Customer Churn Prediction-Telecommunications
Role: Data Analyst(Predictive Modeling)
Deliverable: 3-1(a) Defined Architecture of ANN Model
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1. Purpose

This document specifies the architecture of the Artificial Neural Network (ANN) developed to predict customer churn for a telecommunications company. The model takes a set of customer attributes and behaviours as input and outputs a probability that the customer will churn, enabling targeted retention interventions.

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2. Problem Type

- Task: Binary classification
- Target variable: `Churn` (0 = customer stayed, 1 = customer churned)
- Output: Probability between 0 and 1, thresholded at 0.5 by default to produce a discrete prediction

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3. Input Layer

- Number of neurons: 10
- Input features (one neuron per feature):

#	Feature	Description
1	gender	Customer gender (encoded)
2	SeniorCitizen	Whether the customer is a senior citizen
3	Dependents	Whether the customer has dependents
4	tenure	Number of months the customer has stayed (standardised)
5	PhoneService	Whether the customer has phone service
6	MultipleLines	Whether the customer has multiple lines
7	InternetService	Type of internet service (encoded)
8	MonthlyCharges	Monthly bill amount (standardised)
9	Contract_One_year	Whether the customer is on a one-year contract
10	Contract_Two_year	Whether the customer is on a two-year contract

All inputs are numeric and preprocessed (categorical variables encoded, continuous variables standardised) by the Stage 2 data preparation pipeline.

4. Hidden Layers

The network uses two hidden layers with dropout regularisation between them. This is deliberately compact, appropriate for a 10-feature input and a training set of 5,634 rows, to balance predictive capacity against the risk of overfitting.

Layer	Type	Size	Activation	Purpose
Hidden 1	Dense (fully connected)	16 neurons	ReLU	Learns non-linear interactions between input features
Dropout 1	Dropout	rate = 0.3	-	Randomly deactivates 30% of neurons during each training step to reduce overfitting
Hidden 2	Dense (fully connected)	8 neurons	ReLU	Compresses the learned representation toward the output decision

Rationale for ReLU: ReLU is computationally efficient, avoids vanishing gradients during backpropagation, and is standard for feed-forward classification networks.

Rationale for dropout: With only 5,634 training examples, the network can otherwise memorise the training data. A dropout rate of 0.3 forces the model to learn redundant representations, improving generalisation.

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5. Output Layer

Layer	Type	Size	Activation	Purpose
Output	Dense	1 neuron	Sigmoid	Outputs a churn probability between 0 and 1

Rationale for sigmoid: The sigmoid function maps any real-valued input to the (0, 1) interval, which is interpreted as the probability of the positive class (churn). This is the standard activation for binary classification with a single output neuron.

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6. Model Summary

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text
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Model: "sequential"
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┃ Layer (type)                         ┃ Output Shape                ┃         Param # ┃
┃
┡━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━┩
┃ hidden_1 (Dense)                     ┃ (None, 16)                  ┃             176 ┃
┃ dropout_1 (Dropout)                  ┃ (None, 16)                  ┃               0 ┃
┃ hidden_2 (Dense)                     ┃ (None, 8)                   ┃             136 ┃
┃ output (Dense)                       ┃ (None, 1)                   ┃               9 ┃
Total params: 321 (1.25 KB)
Trainable params: 321 (1.25 KB)
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Non-trainable params: 0 (0.00 B)
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Total trainable parameters: 321.

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7. Compilation and Training Configuration

Setting	Value	Rationale
Optimiser	Adam	Adaptive learning rate; converges quickly on small networks
Loss function	Binary cross-entropy	Standard loss for binary classification
Primary metric	Accuracy	Supplemented by precision, recall, F1, and ROC-AUC during evaluation
Batch size	32	Standard mini-batch size; balances convergence speed and stability
Maximum epochs	50	Upper bound; early stopping typically ends training sooner
Early stopping	patience = 10, monitor = val_loss, restore_best_weights = True	Prevents overfitting; restores the best-performing weights

Validation split	20% of training data	Used only for early stopping, not for final evaluation
Class weights	`{0: 0.68, 1: 1.88}`	Compensates for the 73.5 / 26.5 class imbalance, prioritising recall on the churn class

Training stopped at epoch 33, and the model restored the weights from epoch 23 (lowest validation loss).

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8. Summary of Architecture Choices

- Compact network (321 parameters): appropriate for a 10-feature dataset and reduces overfitting risk.
- Two hidden layers (16 → 8): enough capacity to model non-linear interactions without excessive complexity.
- Dropout (0.3): regularisation tailored to a modest dataset.
- ReLU activations: efficient and avoids vanishing gradients.
- Sigmoid output: produces a well-defined churn probability.
- Class weighting: aligns the model's objective with the business priority, catching churners, even at the cost of some false positives.
- Early stopping: guards against overfitting and ensures the best model is retained.